Modeling Typing Dynamics for Enhanced Password Security: A Statistical Exploration

Introduction

This report documents our exploration of incorporating typing dynamics into a password-based security system, navigating the intricate realm where technology converges with human behavior. Tasked with identifying a lower training threshold based on the repetition of a specific passcode (.tie5Roanl), our group analyzed data from 51 individuals across 8 sessions. Each session consists of 50 repetitions which results into 400 repetitions for one subject. The forthcoming sections will detail our methodological approach and unveil the insights gleaned from users' typing behaviors, offering a glimpse into the potential advancements in biometric security.

Problem Statement

The challenge at hand involves deciphering the intricacies of individual typing behaviors. For every subject in our dataset, the task is to distill a subset of their typing data that accurately represents the acquired typing behavior. This undertaking aims to pinpoint the key characteristics and nuances in the repetitive entry of a specific passcode, paving the way for a refined understanding of learned typing behaviors.

Objective:

The objective of this project is to identify the most suitable independent variable for constructing a linear regression model that, after undergoing False Discovery Rate (FDR) control, maintains a high false positive rate. The primary focus is on selecting a variable that not only contributes meaningfully to the model but also adheres to all assumptions of linear regression. By carefully considering both statistical significance and adherence to regression assumptions, we aim to build a robust model that strikes a balance between predictive power and reliability in the presence of controlled false positives.

Approach 1: *FIND A SUBSET OF THEIR TYPING DATA THAT IS REPRESENTIVIE OF THE LEARNED TYPING BEHAVIOR?*

To systematically capture the intricacies of learned typing behaviors, a covariate was formulated to quantify the historical frequency of passcode entries for each subject. This covariate, number of repetitions, serves as a predictor variable, played a pivotal role in modeling the responses associated with "H, DD, and UD."

The described procedures aim to identify a refined subset of data that is representative of acquired typing behavior. Let's break down each step and provide a statistical interpretation:

- **Find the difference between subsequent datasets:**

This step involves calculating the differences between consecutive data points. Mathematically, if you have a dataset $X= [x_1, x_2, x_3,...,x_n]$X=[x1,x2,x3,...,xn], you calculate differences as $D =[x_2-x_1, x_3-x_2, ..., x_n-x_{n-1}]$

- **Plot the differences against the number of repetitions:**

The differences are then plotted against the number of repetitions. This creates a graphical representation of how the typing behavior changes over repetitions.

- **Observation and selection of points near the zero y-axis (plateau section):**

The focus is on identifying points in the plot that are close to the zero y-axis. This suggests a plateau section where the typing behavior remains relatively stable or unchanged.

- **Discard subjects showing a monotonic increase or no plateau:**

Subjects displaying a monotonic increase (consistent upward trend) or those without a plateau are discarded. This indicates that their typing behavior does not stabilize or show a consistent pattern over repetitions.

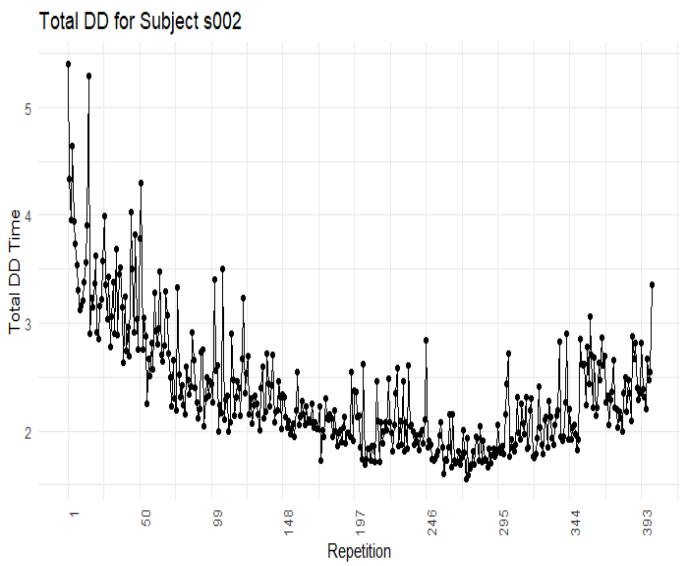
Statistical Interpretation:

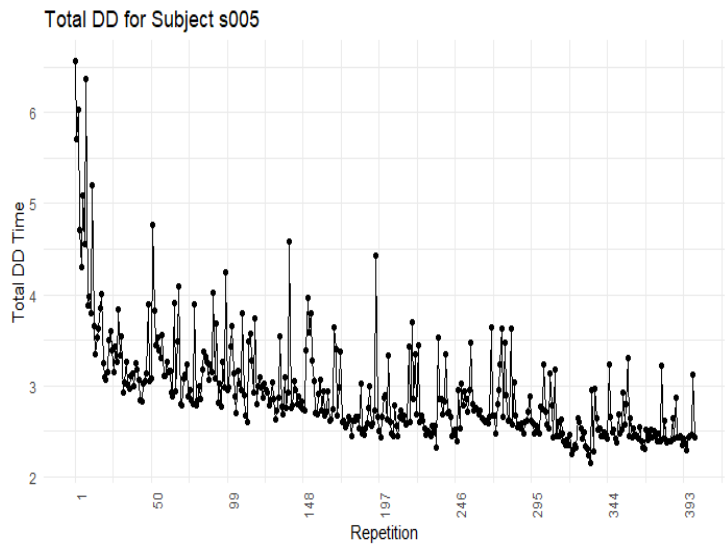
The procedure is essentially looking for stability or a plateau in the typing behavior over repetitions. A plateau suggests that the typing behavior has reached a relatively constant state, which may be indicative of a more refined and consistent typing pattern.

The focus on points near the zero y-axis could be related to identifying portions of the data where the typing behavior is not changing significantly, indicating a stable typing state.

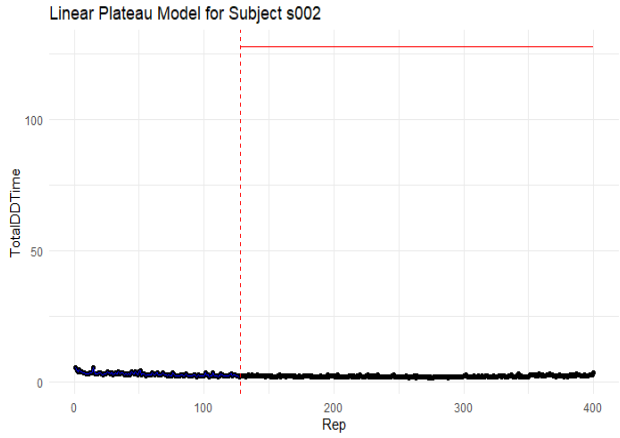
The decision to discard subjects without a plateau or showing a monotonic increase suggests a desire for a subset of data that reflects a stable and consistent typing behavior over time. This can be important for characterizing and analyzing typing patterns in a more controlled and representative manner.

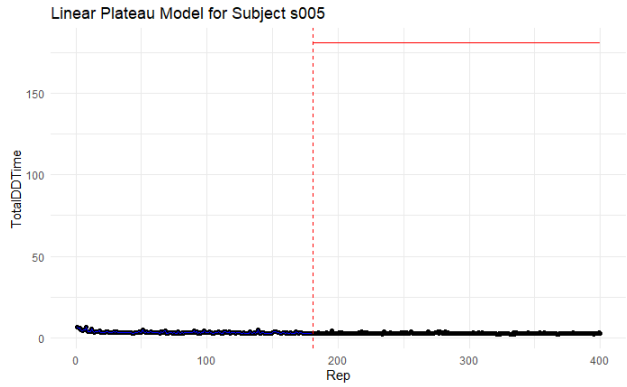
Result and Interpretation from DD Variable:

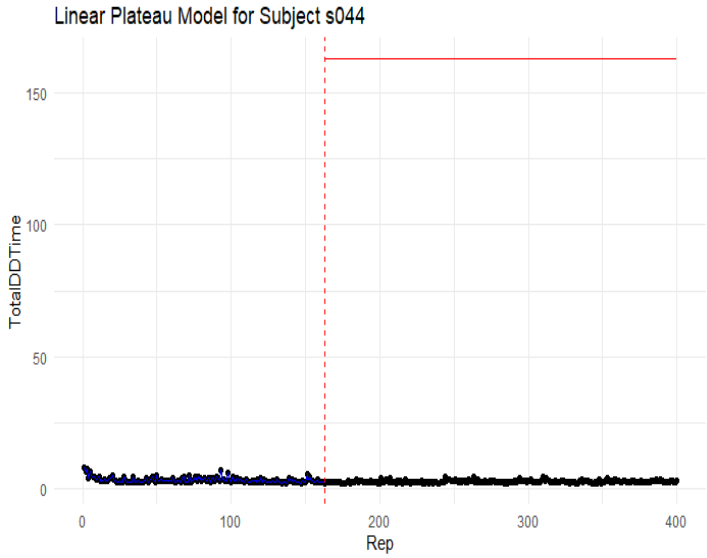


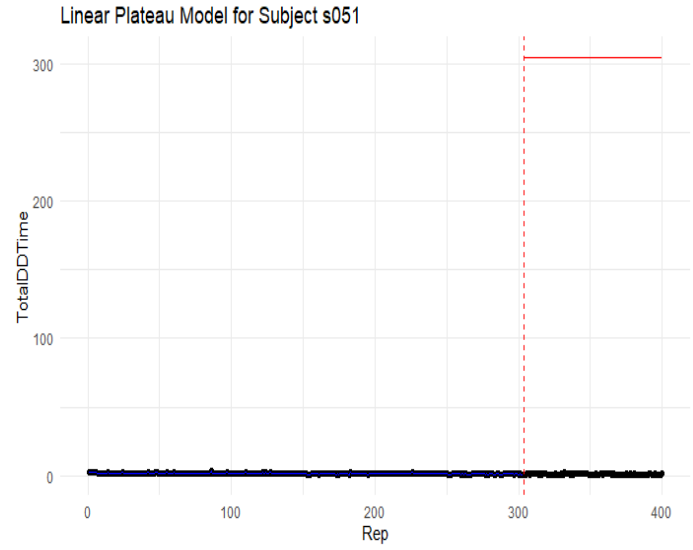


Subjects from the total DD variable show similar pattern and we were able to create subset of stable dataset from the distribution. Further, more we moved on with the **plateau** method to find that cutoff point for each subject.





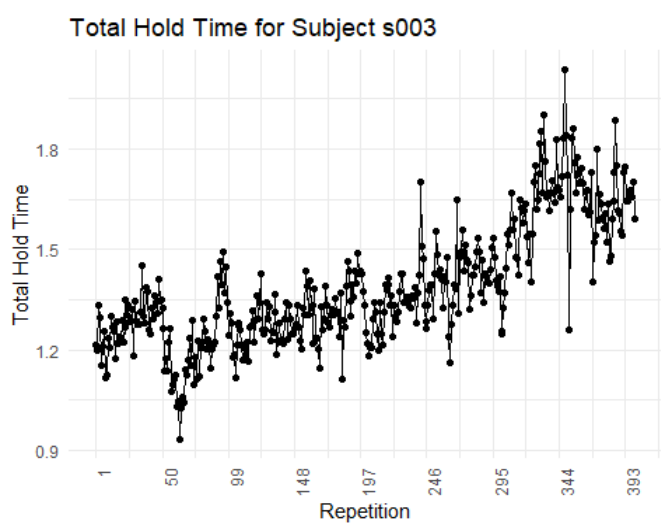




Result and Interpretation from HD Variable

The result and interpretation from the HD (presumably a variable denoted as "HD") variable suggest that a decision was made to drop or exclude this variable from further analysis. Majority of the HD variable shows a monotonic increase in trend as seen in the graph. This statement indicates that when the HD variable is plotted against the number of repetitions, the majority of data points exhibit a consistent and monotonic increasing trend. In other words, the values of the HD variable tend to steadily increase or decrease with each repetition, and there is no apparent plateau or stable section.

Therefore, the decision to drop the HD variable is likely driven by the desire to focus on typing behavior instances that exhibit stability or plateau over repetitions. A monotonic increase in the HD variable suggests a lack of stability or consistent typing behavior and might indicate a variable that does not contribute meaningful information for the analysis being conducted. A monotonic increase might imply that the HD variable is not providing insights into a refined or stable typing pattern, which could be essential for the specific goals of the analysis. The exclusion of the HD variable could be justified by the need to work with a subset of data that better represents stable typing behavior. This decision ensures that the subsequent analysis or modeling efforts are focused on instances where the typing behavior is more consistent and less influenced by systematic increases or decreases in the HD variable.



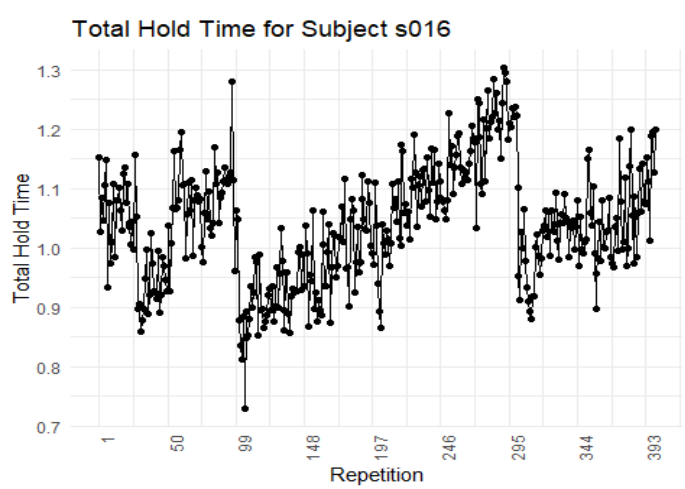


Figure 2a: Monotonic Increase in dataset

In summary, the result and interpretation suggest that the HD variable was dropped because the majority of its values demonstrated a monotonic increasing trend. This trend might indicate a lack of stability in typing behavior, and the decision to exclude the variable reflects a focus on refining the dataset to better capture instances of stable and consistent typing patterns

Assumption of Linearity to justify the removal of total HD time using Residual and Fitted Value

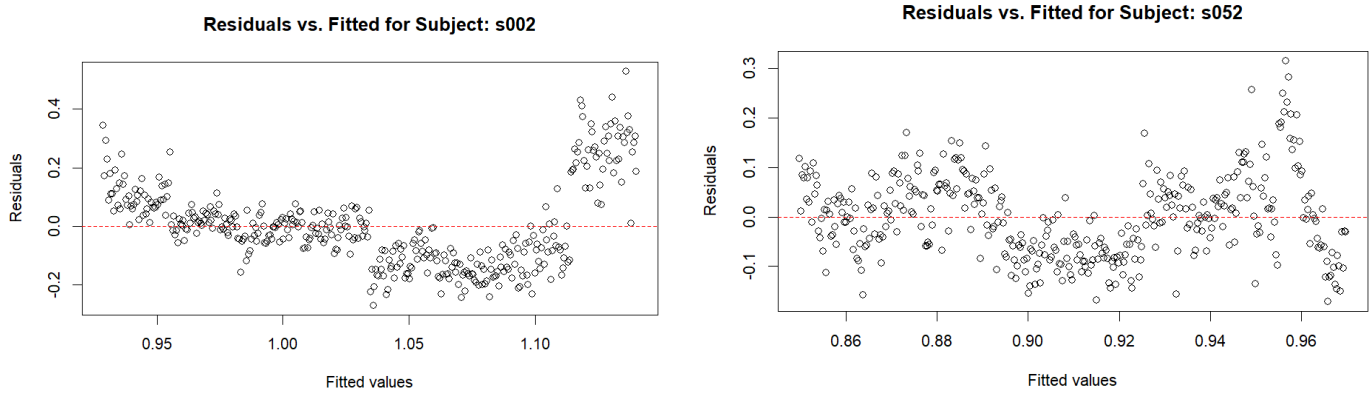


Figure 3: Graph of Residuals against fitted value of total HD time

In justifying the removal of the Total HD time variable, We examined the assumption of linearity through the Residuals vs. Fitted Values plot as seen in figure 3. The analysis revealed that the relationship between Total HD time and the response variable did not align with the assumption of linearity. The plot displayed a discernible pattern, indicating that the linear model failed to capture the underlying structure of the data. Consequently, the decision to exclude Total HD time was grounded in the violation of the linearity assumption, ensuring that the chosen variables adhere to the fundamental assumptions of linear regression.

Result and Interpretation from UD Variable:

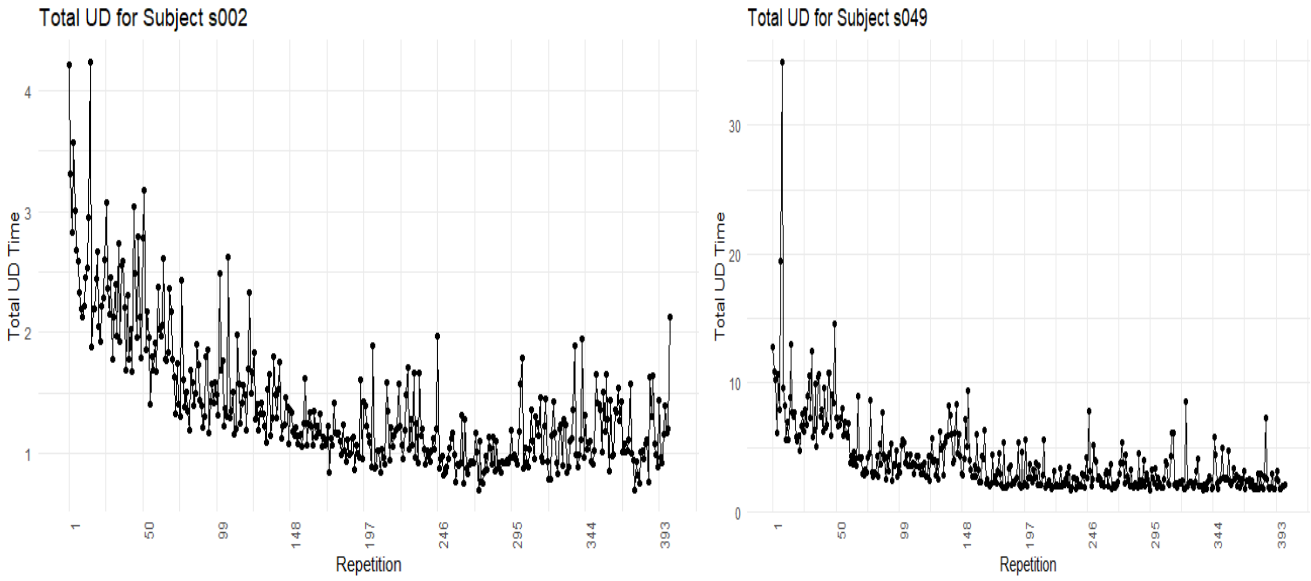
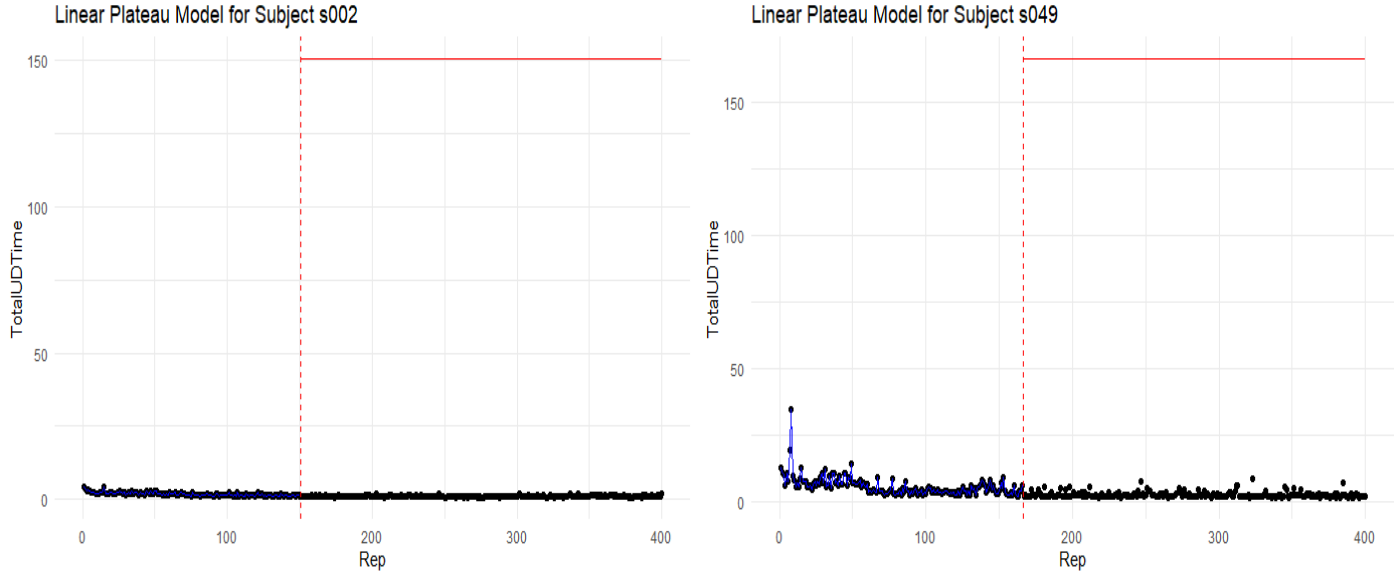
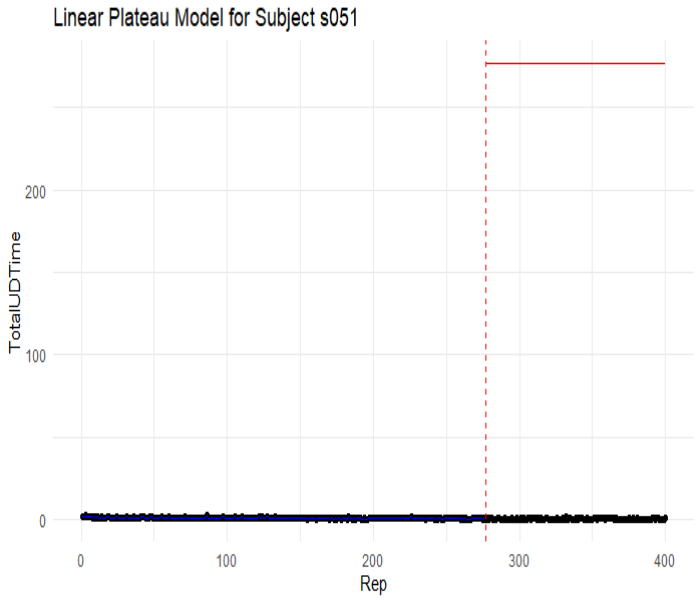
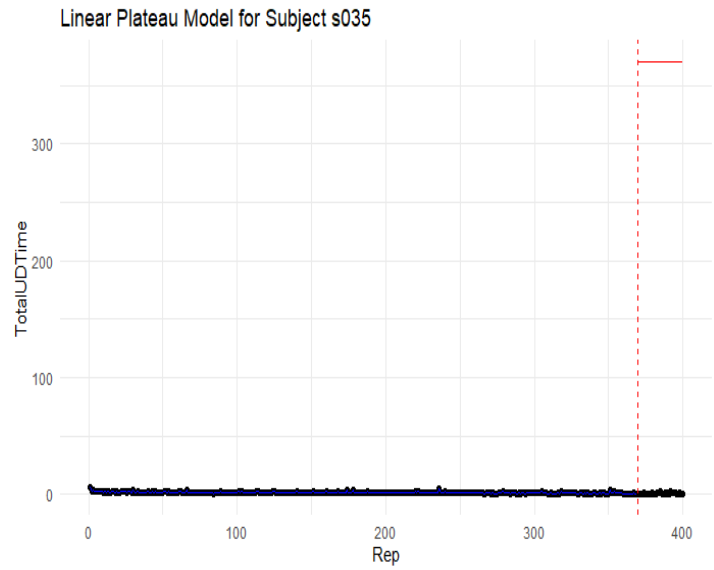


Figure 4: Concentration of dataset around zero axis

In general, we were able to create a subset that suggests that the data within that subset exhibits a certain level of stability or homogeneity in terms of the characteristics being analyzed. Furthermore, we move on to Plateau method to find that cutoff point.







Building a regression models for the total UD, total DD, and total H times

Regression model was constructed at 5% level of significance for each of the response variables (total UD, total DD) with the following Hypothesis:

Null Hypothesis: There is no relationship between the response variable and repetitions

Alternative: There exists a relationship between the response variable and repetitions

$Y = Bo + B_1X + E$

Y: The variable you are trying to predict or explain.

Bo: The intercept, the value of Y when X is 0.

B_1: The slope, the change in Y for a one-unit change in X.

X: The variable that is used to predict Y, often referred to as the independent variable.

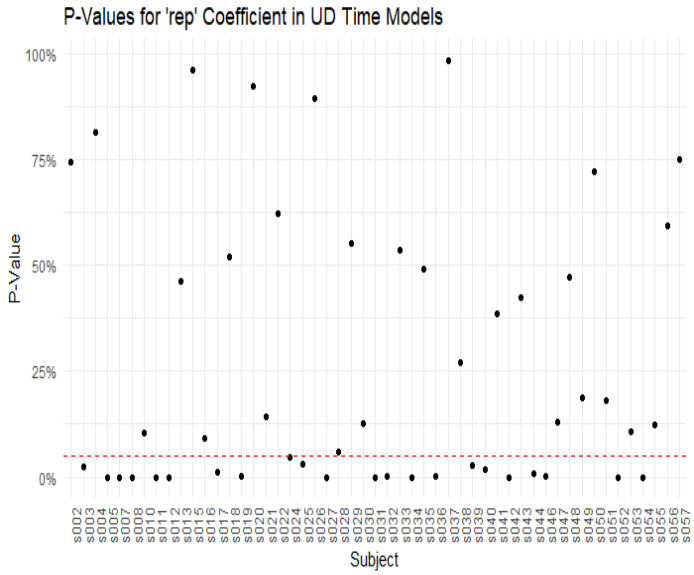
E: The error term, representing the unobserved factors that affect Y but are not included in the model.

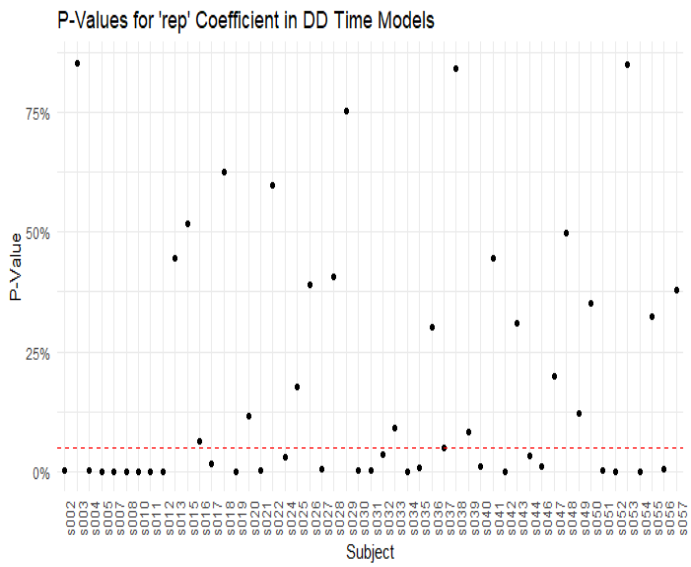
The result of the regression model for each subject in relation to their total UD, total DD can be seen on the Index page.

If p-value < 0.05, we reject the null hypothesis in favor of the alternative hypothesis and if p – value > 0.05, we fail to reject the null hypothesis.

Use FDR's to assess reasonableness of the normality assumptions of the residuals after fitting model for all subjects.

Graph of P values for Total UD and DD Times for each subject subsect

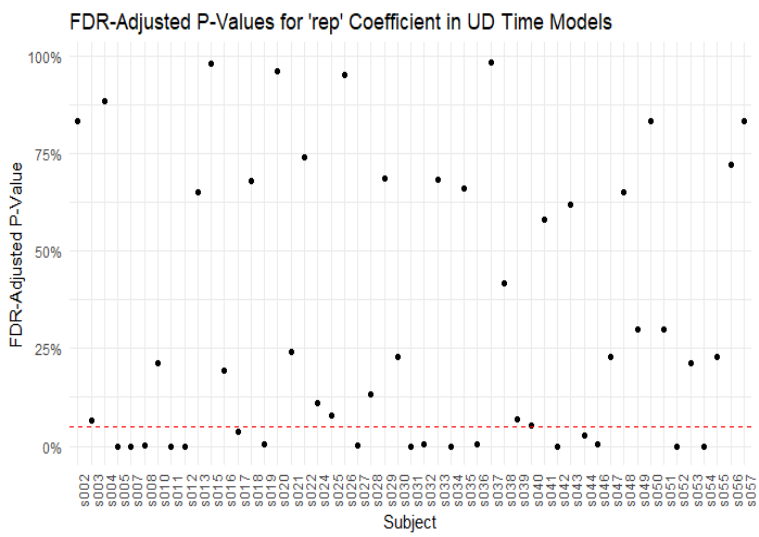


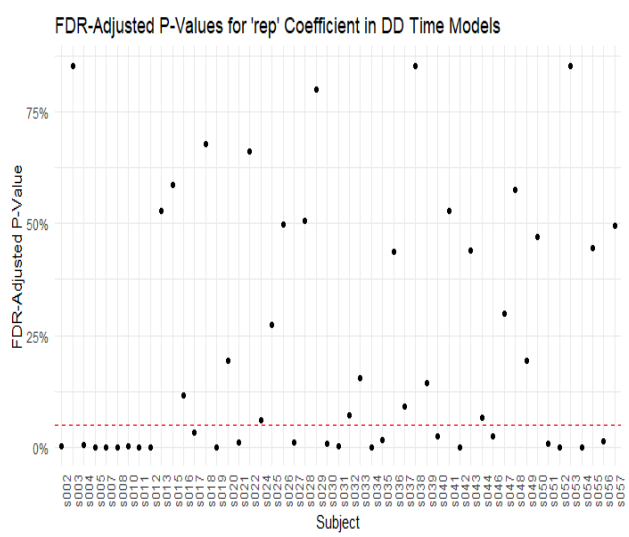


FALSE DISCOVERY RATE

Adjusting the p-value using False Discovery Rate (FDR) was undertaken to mitigate the risk of Type I errors, which involve incorrectly rejecting a true null hypothesis. By applying FDR corrections, we aimed to control the overall rate of false positives (i.e, rejecting the null hypothesis when the null hypothesis is true) in our statistical inferences, ensuring a more robust and reliable interpretation of the relationship between repetition and typing behavior metrics.

FDR – Corrected p-values





Summary of corrected p-values

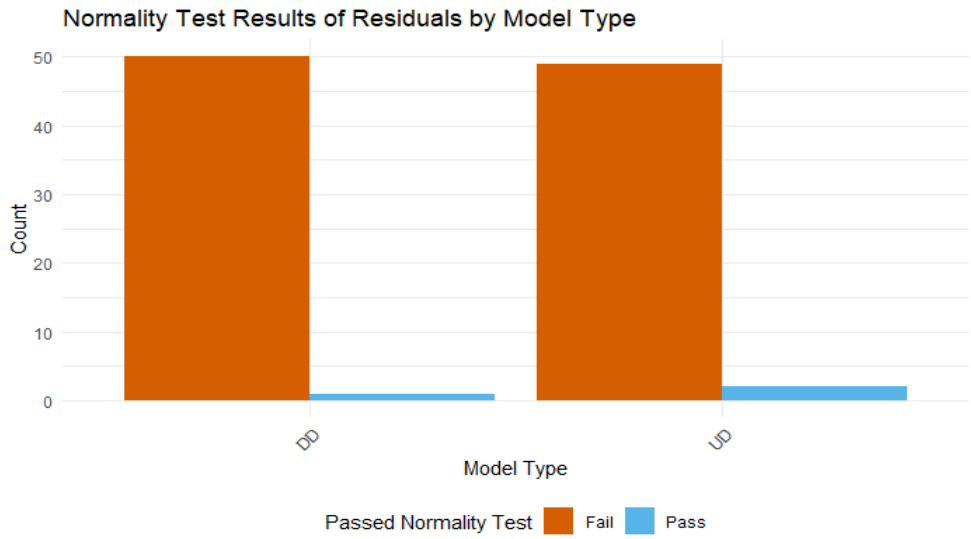
- Total DD time

Number of subjects with significant p-values before FDR correction: 27
Number of subjects with significant p-values after FDR correction:23

- **Total UD time**

Number of subjects with significant p-values before FDR correction: 22
Number of subjects with significant p-values after FDR correction: 17

Normality Test for Residuals.
We conducted normality tests on the residuals of each model to draw inferences about the estimated coefficients using shapiro wilk test, and the results are presented below.



Total UD time - Subjects with significant p-values after FDR

Subject	FDR_P_Value
s005	6.728059e-10
s007	1.212796e-05
s008	2.421441e-04
s011	3.624318e-35
s012	1.538249e-08
s017	3.554861e-02
s019	3.732639e-03
s027	6.890512e-04
s031	1.348531e-05
s032	4.042088e-03
s034	3.758017e-06
s036	3.690102e-03
s042	1.348531e-05
s044	2.829746e-02
s046	5.931785e-03
s052	2.595681e-07
s054	6.728059e-10

Total DD time - Subjects with significant p-values after FDR

Subject	FDR_P_Value
s002	1.527793e-03
s004	3.676348e-03
s005	1.009360e-05
s007	4.479103e-05
s008	1.628530e-05
s010	3.924132e-04
s011	1.477395e-22
s012	3.309647e-14
s017	3.188044e-02
s019	5.254941e-05
s021	8.748050e-03
s027	8.874922e-03
s030	7.396241e-03
s031	7.970025e-04
s034	3.872653e-05
s035	1.478706e-02
s040	2.281086e-02
s042	4.283590e-07
s046	2.461574e-02
s051	7.171387e-03
s052	5.358899e-06
s054	9.351888e-07
s056	1.161513e-02

The sample size for each subject, $n > 30$, from central limit theorem, we can conclude that how subset is normally distributed. Although, the highlighted subjects show normal distribution from shapiro wilk test ($p > 0.05$)

Why Total UD was Chosen:

Since DD is the sum of UD and HT. We would recommend UD for following reasons:

- **High Retention of Significance after FDR:**

Total UD was chosen as the preferred variable due to its higher retention of statistically significant associations even after applying False Discovery Rate (FDR) correction.

- **Enhanced Robustness:**

The elevated number of p-values below the 0.05 threshold for Total UD indicates a more robust association with typing dynamics, aligning with the objective of minimizing false positives while maintaining statistical significance.

- **Strategic Control Over False Positives:**

Opting for Total UD aligns with a strategic decision to prioritize variables that withstand a more stringent control over the rate of false positives, ensuring the reliability and validity of the identified associations.

Conclusion and Recommendation

Given that Total DD Time is composed of both Up-Down Time and Hold Time, there is a possibility that the overlapping effects between these components could influence the observed relationships. Therefore, Total UD Time emerges as a more distinct and reliable indicator for assessing the correlation with the number of password repetitions.

In conclusion, the analysis supports the robustness of the relationship between the number of password repetitions and Total UD Time. Based on these findings, it is recommended to prioritize Total UD Time as the key response variable for further studies or applications that aim to understand or utilize typing dynamics in the context of repeated password entries. This focus on Total UD Time is advocated due to its clearer statistical significance and lower likelihood of being affected by overlapping time components compared to Total DD Time.

Statistical Insight:

The FDR-adjusted p-values signify that, after accounting for multiple testing, the evidence supporting a relationship between the number of password repetitions and Total UD remains statistically significant.

Recommendation:

Given the FDR-corrected findings, it is recommended to prioritize further investigation and consideration of the relationship between the number of password repetitions and Total UD. This specific aspect demonstrates a more robust and reliable statistical significance, warranting deeper exploration and potential implications for understanding the dynamics of typing behavior. For Total HD, caution is advised as the dataset shows a monotonic increase or wide variations within the dataset.

Reasons for selecting Plateau method over sequential differencing.

In our study, the Plateau Method demonstrated superior efficiency compared to the Sequential Differences technique when subsetting a dataset to obtain a stable dataset. The choice of the Plateau Method was driven by several factors, each contributing to its effectiveness in our specific context:

Adaptive Thresholding for Robust Selection:

- The Plateau Method employs an adaptive thresholding approach, which proved beneficial in our dataset. This adaptive nature allowed the method to dynamically adjust to the varying characteristics of the data, ensuring robust feature selection that was not overly influenced by noise or short-term fluctuations.

Resilience to Local Variations:

- The Plateau Method exhibited resilience to local variations and minor fluctuations in the dataset. By focusing on identifying plateaus, it successfully disregarded short-term changes and localized extrema, resulting in a more stable subset that better represented the underlying trends.

Global Optimum Consideration:

- Our objective was to identify a stable dataset that encapsulates global trends rather than being sensitive to local changes. The Plateau Method's emphasis on finding plateaus allowed it to prioritize stable regions of the evaluation metric, avoiding overfitting to specific segments of the data.

Effective Handling of Plateaus and Stability:

- The Plateau Method's inherent design, specifically tailored for stability, proved effective in handling plateaus. This capability ensured that the selected subset represented periods of stability in the data, aligning with our goal of obtaining a stable dataset.

Adaptability to Data Distribution:

- The Plateau Method's adaptive thresholding and global optimum consideration showcased its adaptability to the distribution of our dataset. This adaptability contributed to its effectiveness in identifying and preserving stable periods in the data.

In summary, the Plateau Method emerged as the preferred technique for subsetting our dataset to achieve stability. Its adaptive nature, resilience to local variations, focus on global optimums, effective handling of plateaus, and adaptability to data distribution collectively made it a more efficient and suitable approach for our specific feature selection requirements. The number of False positive increased for both UD and DD after the FDR adjusted p-values.

Refrence

- https://www.cs.cmu.edu/~maxion/pubs/KillourhyMaxion09.pdf
- Jafari, M., & Ansari-Pour, N. (2019). Why, when and how to adjust your P values?. *Cell Journal (Yakhteh)*, *20*(4), 604.
- van Iterson, M., Boer, J. M., & Menezes, R. X. (2010). Filtering, FDR and power. *BMC bioinformatics*, *11*(1), 1-11.